MANIT GUPTA

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EDUCATION

University of Louisville — J.B. Speed School of Engineering ***Expected Graduation: May 2029***

B.S. Computer Science (AI/ML Concentration) | Minors: Mathematics, Industrial & Systems Engineering | GPA: 4.0/4.0

- Junior standing via 16 AP/dual-credit courses; Henry Vogt Scholarship ($64,000); Valedictorian, duPont Manual High School

Current Coursework: Discrete Structures, Engineering Analysis III (Calc), Decision Support Systems

TECHNICAL SKILLS

Languages: Python, R, Java, C++, C, JavaScript, SQL
AI Safety & Evaluation: Conformal Risk Control, Clopper-Pearson Bounds, Pre-Registered Analysis, Nested CV, Permutation Testing
ML / Computer Vision: PyTorch, Quantized VLMs, llama.cpp, LightGBM, OpenCV, Pandas
Frameworks, Cloud & Hardware: Django, Flask, Node.js, Git, CI/CD, GCP Vertex AI, Azure, PostgreSQL, Raspberry Pi, Jetson

RESEARCH EXPERIENCE

Independent Researcher — H-CASES: Certified Escalation on a $100 Edge Device ***2026 – Present***

Mentor: Dr. Luis Segura, PhD, UofL | Research letter in preparation (IEEE target) | KJAS poster accepted | Patent developing

- Built a conformal risk-control system running a quantized (8-bit) InternVL3-1B VLM offline on a $100 Raspberry Pi 5 to decide which cases to auto-clear vs. route to a human
- Pre-registered Clopper-Pearson miss-rate thresholds before scoring 368 held-out cases across 12 camera sites; met all pre-registered safety targets, incl. a site-shift test
- Diagnosed a silent on-device serving bug with a larger effect than 8- vs 16-bit quantization; numbers re-derived by a 373-test code audit

Technical Lead — Gut Microbiome–Autism Multi-Omics Study ***2026 – Present***

University of Louisville | PI: Dr. Naveen Nagarajan | 10-person team | Manuscript in preparation (Nature-level target)

- Architected a family-disjoint, leak-free ML pipeline for 819 children (773 families) across 2,179+ species and 548 pathways
- Pre-registered 7 test families (107,077 comparisons); quantified how data leakage inflates apparent findings
- Cut team development time 50%+ via CI/CD; Co-Lead of a PROSPERO-registered review (CRD420261470724), directing 5 students

First Author — Real-Time PPE Compliance & Clinical Tool Monitoring Using Generative AI ***2024 – 2025***

Published, Cureus Journal of Medicine (PubMed-indexed, PMID: 41280963) | Mentor: Dr. Shuchita Garg | Patent developing

- Achieved 95.8% accuracy and 100% specificity across 168 trials with a real-time Generative AI + OpenCV pipeline

PrenylFinder & Proteomics — 3 Posters, Research!Louisville 2026 ***2025 – 2026***

Mentor: Dr. Mark Running, PhD, UofL Dept. of Biology | First Author, Second Author, and Co-Author posters

- Built PrenylFinder, a browser-native scanner that screened ~95,000 proteins, surfacing 1,500+ candidates at 35–51× enrichment
- Recovered 7/7 canonical Rab GTPases; raised recall 68% → 84.7%; co-authored an Alzheimer's/ALS signature study

TECHNICAL PROJECTS

EyeNova — Assistive Edge-AI Vision Wearable (Co-Founder) ***2025 – Present***

- Delivered a $30–$100 wearable (vs. $5,000+ alternatives) at >95% voice-command accuracy with on-device CV + Gemini API

Diabetes Management App — Congressional App Challenge 3rd; WDRB News ***2025***

- Forecasted hypo/hyperglycemic events at 93–95% accuracy with LightGBM (R^2 = 0.83) on fused CGM/insulin data

Natural-Language Text-to-Rule Surveillance ***2025 – 2026***

- Converted safety documents into machine-enforceable rules via a quantized LLM on Raspberry Pi; >90% detection accuracy

ISKCON Louisville — Official Website & Digital Outreach Platform ***9+ Years***

- Built and deployed a full-stack site from scratch: 163 distinct devices and 487 feature-tab opens in month one

LEADERSHIP & ACHIEVEMENTS

- **Founding Member & Co-Lead Systems Architect, Corsaryn Recovery**; NSF SBIR/STTR pitch submitted
- **Local Lead**, NASA Space Apps Louisville • **Music Director**, ISKCON (3,000+ audience) • **Exec. VP**, Mango Care Foundation
- Henry Vogt Scholar • Governor's Scholar • Broadcom MASTERS National Qualifier • Science Fair Regional 1st / State 3rd
- TSA National Finalist (Webmaster 6th, Debate 10th) • 2× State Debate Champion • Taekwondo Black Belt • Trilingual